

St. Catherine Medical Center — Emergency Department

2100 S. Western Ave, Chicago, IL 60608 | Level II Trauma Center
Phone: (312) 555-0700 | Fax: (312) 555-0701

Patient: Carlos E. Reyes

Date: 01/25/2026

DOB: 10/05/1988

Provider: Patricia Eng, MD

MRN: SCM-0339147

Specialty: Emergency Medicine

CHIEF COMPLAINT

Witnessed seizure — brought in by girlfriend.

HISTORY OF PRESENT ILLNESS

Mr. Reyes is a 37-year-old male with no known medical history who presents after a witnessed generalized tonic-clonic seizure at home. Per girlfriend: He was watching TV when he suddenly let out a cry, went stiff, then started shaking rhythmically in all four limbs. She estimates the convulsion lasted 2 minutes. He was unresponsive during the event. She rolled him onto his side. He was incontinent of urine. After the shaking stopped he was confused and combative for approximately 15 minutes, then became very sleepy. She called 911. He bit the right side of his tongue.

Patient himself has no recall of the event. He says he felt 'fine' before it happened. He denies any prodrome of aura, smell, or visual changes. He denies prior episodes. He does report poor sleep this past week due to work stress and consumed 4-5 beers the night before. He denies illicit drug use. He smokes half a pack per day. No family history of epilepsy. Takes no medications. No recent head trauma or illness. No prior CT or MRI of the brain.

PHYSICAL EXAM

Vitals (on arrival): BP 148/92, HR 110, Temp 37.2°C, O2 sat 98% RA, RR 16

General: Drowsy but arousable, oriented to self only initially. Improved to fully oriented x4 over 45 minutes.

HEENT: Healing laceration right lateral tongue. No head trauma signs. PERRL, 3mm and reactive bilaterally.

Cardiovascular: Tachycardic, regular. No murmurs.

Neuro (post-ictal, on arrival): Drowsy, confused. No focal motor deficits. Moving all four extremities symmetrically. No gaze deviation. No nystagmus.

Neuro (2 hours post-ictally): Alert and oriented x4. Cranial nerves intact. Motor 5/5 all extremities. Sensation intact. Reflexes 2+ and symmetric. Gait normal.

ASSESSMENT AND PLAN

1. First unprovoked generalized tonic-clonic seizure (R56.9)

Classic witnessed GTC seizure with tongue bite, urinary incontinence, and postictal confusion. No clear provoking cause identified, though sleep deprivation and alcohol use are potential lowering factors. Neurological exam normal post-ictally.

Workup:

- CT head without contrast: obtained — results documented separately
- BMP, CBC, glucose, LFTs, UDS, EtOH level: obtained — see labs
- EEG: ordered as outpatient — see referral
- MRI brain with/without contrast: ordered as outpatient

- Counseled patient regarding seizure precautions: no driving, no swimming alone, no operating heavy machinery until cleared by neurology
- Strongly advised alcohol reduction
- No antiepileptic medication initiated — deferring to outpatient Neurology
- Neurology referral placed; requesting appointment within 1-2 weeks
- Return precautions given: return to ED for any recurrent episode

Disposition: Discharged home with girlfriend after 4 hours of observation. Patient fully oriented at discharge, ambulating independently.

Electronically signed by: Patricia Eng, MD
Signed 01/25/2026 22:40 CST | St. Catherine ED EHR | Attending: Marcus Webb, MD

University Radiology Group — St. Catherine Medical Center

2100 S. Western Ave, Chicago, IL 60608
Phone: (312) 555-0750 | Fax: (312) 555-0751

RADIOLOGY REPORT

Patient: Carlos E. Reyes **DOB:** 10/05/1988 **MRN:** SCM-0339147
Exam Date: 01/25/2026 21:15 **Ordering MD:** Patricia Eng, MD **Accession:** SCM-R-20260125-4471

EXAM

CT HEAD WITHOUT CONTRAST

CLINICAL INDICATION

37-year-old male with first-time generalized seizure. Rule out intracranial pathology.

TECHNIQUE

Axial CT of the head performed without IV contrast using standard head protocol. 5 mm and 1.25 mm reconstructions in axial, coronal, and sagittal planes reviewed.

COMPARISON

None available.

FINDINGS

Brain parenchyma: No acute intracranial hemorrhage. No area of abnormal density to suggest infarction or contusion. Gray-white matter differentiation is preserved. No mass effect or midline shift.

Ventricles/Cisterns: Ventricles are normal in size and configuration. Basal cisterns are patent. No hydrocephalus.

Sulci/Cortex: Normal cortical sulcal pattern for age. No gyral swelling.

Posterior fossa: Cerebellum, brainstem, and fourth ventricle appear unremarkable.

Calvarium/Skull base: No fracture. No bony destructive lesion.

Orbits/Paranasal sinuses: Mild mucosal thickening in bilateral maxillary sinuses, likely chronic/incidental. Orbits grossly unremarkable.

IMPRESSION

1. No acute intracranial hemorrhage, mass, or structural abnormality identified.
2. No imaging evidence of acute infarction on non-contrast CT.
3. Mild bilateral maxillary sinus mucosal thickening — incidental, clinically insignificant.

Note: A normal non-contrast CT does not exclude subtle structural lesions (e.g., cortical dysplasia, low-grade neoplasm). MRI brain is recommended for comprehensive evaluation of new-onset seizure.

Electronically attested by: Yuki Tanaka, MD, Diagnostic Radiology
Read and attested 01/25/2026 23:05 CST | Accession SCM-R-20260125-4471

St. Catherine Medical Center — Laboratory Services

2100 S. Western Ave, Chicago, IL 60608 | CLIA #14D0088440
Phone: (312) 555-0760 | Fax: (312) 555-0761

LABORATORY REPORT

Patient: Carlos E. Reyes **DOB:** 10/05/1988 **MRN:** SCM-0339147
Exam Date: 01/25/2026 21:30 **Ordering MD:** Patricia Eng, MD **Accession:** LAB-20260125-7821

COLLECTION TIME: 01/25/2026 21:30 | **REPORTED:** 01/25/2026 22:55

BASIC METABOLIC PANEL

Test	Result	Reference Range	Flag
Sodium	138	136-145 mEq/L	
Potassium	3.9	3.5-5.0 mEq/L	
Chloride	101	98-107 mEq/L	
CO2 (Bicarbonate)	22	22-29 mEq/L	
BUN	14	7-20 mg/dL	
Creatinine	0.9	0.7-1.3 mg/dL	
eGFR (CKD-EPI)	>60	≥60 mL/min/1.73m ²	
Glucose	94	70-99 mg/dL	
Calcium	9.2	8.5-10.2 mg/dL	

COMPLETE BLOOD COUNT

Test	Result	Reference Range	Flag
WBC	8.4	4.5-11.0 K/μL	
RBC	5.1	4.5-5.9 M/μL	
Hemoglobin	15.2	13.5-17.5 g/dL	
Hematocrit	44.8	41-53%	
MCV	88	80-100 fL	
Platelets	198	150-400 K/μL	
Neutrophils %	72	40-75%	
Lymphocytes %	22	20-45%	

LIVER FUNCTION / OTHER

Test	Result	Reference Range	Flag
AST	28	10-40 U/L	
ALT	32	7-56 U/L	
Total Bilirubin	0.8	0.2-1.2 mg/dL	
Alkaline Phosphatase	74	44-147 U/L	
Total Protein	7.2	6.3-8.2 g/dL	
Albumin	4.3	3.5-5.0 g/dL	
Magnesium	1.9	1.7-2.2 mg/dL	
Phosphorus	3.4	2.5-4.5 mg/dL	

TOXICOLOGY / ALCOHOL

Test	Result	Reference Range	Flag
Serum Ethanol	62	<10 mg/dL (negative)	H
Urine Drug Screen — Cannabinoids	POSITIVE	Negative	H
Urine Drug Screen — Cocaine	Negative	Negative	
Urine Drug Screen — Opiates	Negative	Negative	
Urine Drug Screen — Amphetamines	Negative	Negative	
Urine Drug Screen — Benzodiazepines	Negative	Negative	

Serum ethanol level of 62 mg/dL is above the legal driving limit (80 mg/dL in IL). Clinical correlation advised. Cannabis detected — may lower seizure threshold during withdrawal in chronic users, though causal role uncertain.

Electronically attested by: St. Catherine Laboratory Services, CLIA Certified
 Director: Howard Park, MD, PhD | Reported 01/25/2026 22:55 CST

Chicago Neurophysiology Associates

875 N. Michigan Ave, Suite 1900, Chicago, IL 60611
Phone: (312) 555-0820 | Fax: (312) 555-0821

ELECTROENCEPHALOGRAPHY (EEG) REPORT

Patient: Carlos E. Reyes **DOB:** 10/05/1988 **MRN:** SCM-0339147
Exam Date: 02/03/2026 09:15 **Ordering MD:** Marcus Webb, MD (Neurology) **Accession:** CNA-EEG-20260203-0882

CLINICAL INDICATION

37-year-old male with first unprovoked generalized tonic-clonic seizure on 01/25/2026. Evaluate for epileptiform activity. Prior CT head normal.

RECORDING DETAILS

Standard 21-electrode placement per modified 10-20 international system. Recording duration: 40 minutes. Includes wakefulness, drowsiness, and stage N1/N2 sleep. Hyperventilation performed x3 minutes. Photoc stimulation at 1-30 Hz performed. Video co-recording: yes. Patient cooperative.

BACKGROUND ACTIVITY

Waking background: Posterior dominant rhythm (PDR) of 10 Hz, well-formed, symmetric, and reactive to eye opening. Amplitude 30-50 μ V. Appropriate fronto-central beta activity. No focal slowing. No asymmetry.

SLEEP

Patient transitioned to drowsiness and stage N1/N2 sleep. Vertex waves and sleep spindles identified bilaterally and symmetrically. K-complexes present. No activation of epileptiform discharges during sleep.

INTERICTAL FINDINGS

Intermittent right temporal sharp waves with phase reversal at T4/T6, occurring in brief runs of 2-4 discharges at approximately 2 Hz. Maximum amplitude 80 μ V. These were most prominent during drowsiness and light sleep. Frequency: 6-8 complexes per 10-minute epoch during drowsiness. No generalized spike-wave complexes observed. No continuous focal slowing.

HYPERVENTILATION

Hyperventilation produced mild diffuse slowing, normal response. No epileptiform activation.

PHOTIC STIMULATION

No photoparoxysmal response at any flash frequency.

NO CLINICAL SEIZURE

No clinical or electrographic seizure was recorded during this study.

IMPRESSION

1. Mildly ABNORMAL EEG.
2. Intermittent right temporal sharp waves during drowsiness and sleep are epileptiform and increase the risk of recurrent seizure. This finding is consistent with a focal epileptic tendency, though the index event was clinically generalized — secondary generalization from a right temporal focus is possible.
3. No generalized epileptiform discharges identified.
4. Normal background activity.

Clinical correlation required. MRI brain with epilepsy protocol is recommended to evaluate for structural etiology in the right temporal region (e.g., mesial temporal sclerosis, cortical dysplasia, neoplasm).